

APPENDIX D TABLE: LAB MONITORING FOR PSYCHIATRIC MEDICATIONS

This is a short and sweet table listing the medications that most psychiatrists would agree require lab monitoring. Our recommendations are quite abbreviated, and we haven't spelled out whether you should order labs before or after starting the medications, nor how you should do follow-up monitoring. There's just too much variation in practice for us to give authoritative detailed guidelines. These medications are mainly here to jog your memory, so you don't forget to at least consider what type of monitoring to do.

Medications	Recommended Laboratory Tests
Acamprosate	BUN/creatinine if renal impairment is suspected
Amantadine	BUN/creatinine if renal impairment is suspected
Antipsychotics—second generation, primarily clozapine, olanzapine, quetiapine, paliperidone, risperidone ¹	Fasting glucose and lipids
Atomoxetine	LFTs
Carbamazepine	CBZ level, complete blood count (CBC), sodium, LFTs, pregnancy test, HLA-B*1502 in Asians ²
Chlorpromazine	ECG if cardiac disease
Citalopram	ECG if cardiac disease, dose ≥ 40 mg/day
Clozapine	Fasting glucose and lipids, CBC
Desvenlafaxine	Periodic BP
Deutetrabenazine	ECG if cardiac disease
Disulfiram	LFTs if liver disease is suspected
Duloxetine	LFTs if liver disease is suspected ³
Gabapentin	BUN/creatinine if renal impairment is suspected
Levomilnacipran	Periodic BP/pulse rate
Lithium	Li level, TSH, BUN/creatinine ⁴ , pregnancy test, ECG if cardiac disease
Methadone	ECG if cardiac disease
Mirtazapine	Lipids
Naltrexone	LFTs if liver disease is suspected
Oxcarbazepine	Sodium, HLA-B*1502 in Asians ²
Paliperidone	Prolactin if symptoms, fasting glucose and lipids
Pregabalin	BUN/creatinine if renal impairment is suspected
Risperidone	Prolactin if symptoms, fasting glucose and lipids
SSRIs	Sodium in elderly if fatigue, dizziness, confusion
Stimulants	ECG if cardiac disease
Thioridazine	ECG if cardiac disease
Topiramate	Bicarbonate
Tricyclic antidepressants	ECG if cardiac disease
Valbenazine	ECG if cardiac disease
Valproic acid	VPA level, LFTs, CBC for platelets, pregnancy test, ammonia if confusion
Venlafaxine	Periodic BP
Ziprasidone	ECG if cardiac disease

¹ Some guidelines recommend monitoring glucose and lipids with all SGAs.

² HLA-B*1502 is a gene that increases the risk of developing toxic epidermal necrolysis (TEN) and Stevens-Johnson syndrome (SJS) in response to taking carbamazepine. Asians, especially the Han Chinese, are much more likely to have the gene than other populations.

³ Duloxetine should not be prescribed in patients with significant alcohol use or evidence of chronic liver disease as it can lead to hepatic failure in rare cases (Cymbalta prescribing information). While the manufacturer does not recommend baseline LFTs for all patients, some clinicians do so anyway to be extra cautious.

⁴ The serum creatinine is used to compute the estimated glomerular filtration rate (eGFR), a more precise measure of kidney functioning. Increasingly, laboratory test results include the estimated GFR. You can calculate it yourself using an online calculator at <https://www.niddk.nih.gov/health-information/communication-programs/pkdep/laboratory-evaluation/glomerular-filtration-rate-calculators>.

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